Xiaowei Ou

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Education

Ph.D. in Physics, Yale University Sep. 2021 – present
Advisor: Prof. Vidvuds Ozoliņš.
Research: neural-network quantum states and variational Monte Carlo for strongly correlated systems.

B.S. in Physics, University of Science and Technology of China (USTC) Aug. 2017 – Jun. 2021

Skills

Programming: Python, C/C++, Bash
ML/Computing: JAX, PyTorch, TensorFlow, NetKet, GPU/HPC (SLURM), Git, Linux
Methods: Monte Carlo, stochastic optimization, numerical linear algebra

Research Experience

Neural Quantum States and Optimization for the Quantum Sign Problem Sep. 2022 – Sep. 2024

- Diagnosed why standard stochastic reconfiguration fails on frustrated magnets: the phase network trains much more slowly than the amplitude network, so Monte Carlo samples collapse and optimization stalls in a local minimum.
- Developed a modified stochastic reconfiguration (natural-gradient) optimizer that evolves the amplitude and phase networks with separate step sizes, learning complex wavefunctions with no pretraining or any imposed sign priors.
- Reached energies within 0.01% of the exact value at $J_2/J_1 = 0$ and 0.7% at the maximally frustrated point $J_2/J_1 = 0.5$ on a periodic 6×6 lattice, using only 2,745 parameters; open-sourced an end-to-end TensorFlow VMC framework.

Deep Neural Backflow for Strongly Correlated Electrons Sep. 2023 – Sep. 2025

- Co-developed a multi-determinant neural backflow wavefunction for the 2D Hubbard model, exploiting low-rank structure shared across determinants for $\sim 10\times$ faster training iterations (on a 4×16 lattice with 256 determinants).
- Reached ground-state energies within 0.45% of auxiliary-field quantum Monte Carlo on lattices up to 10×10 at half filling, and reproduced the intertwined charge and spin stripe order at $1/8$ hole doping found by other methods.
- Built diagnostics (Gram matrices, singular-value spectra, PCA) showing that a deeper backflow network, more than adding determinants, improves accuracy and captures the sign structure with fewer dominant principal components.

Electron–Phonon Coupled Systems with Neural Quantum States Jan. 2025 – Present

- Extended neural quantum states to coupled fermion–boson systems (2D Su–Schrieffer–Heeger model, bond-coupled polarons), where standard neural-network wavefunctions struggle to capture the effects of electron–phonon coupling.
- Ran parameter sweeps over phonon frequency and electron–phonon coupling strength in bond-coupled polaron models, benchmarking our variational energies against published results together with collaborators at the Flatiron Institute.

Publications

- **X. Ou**, T. Huang, V. Ozoliņš, "Improving neural network performance for solving quantum sign structure," *Phys. Rev. B* **112**, 165122 (2025).
- T. Huang, **X. Ou**, V. Ozoliņš, "Neural Network Backflow with Low-Rank Multi-Determinant Updates," arXiv:2609.26544 (2026), submitted to *Phys. Rev. B*.

Selected Presentations

- APS March Meeting 2023 (Las Vegas) and 2024 (Minneapolis); APS Global Physics Summit 2026 (Denver) – oral.
- Annual Yale Bouchet Conference on Graduate Education 2026 – oral.
- Physics of John Hopfield: Learning and Intelligence Workshop 2025 (Princeton); Yale Energy Sciences Institute Retreat 2024 – poster.

Awards & Honors

- Yale Graduate Student Assembly Conference Travel Award (2025)
- Yale Energy Sciences Institute Retreat, 2nd Place Poster Award (2024)
- USTC Funding for Excellent Undergraduates in Fundamental Science (2018 – 2021)

Leadership & Outreach

- Co-organizer, *Building AI Agents from Scratch* workshop series, Yale (Fall 2026); *Building LLMs from Scratch* workshop series, WTI Center for Neurocomputation and Machine Intelligence, Yale (Spring 2026).
- Co-organizer, Yale Physics Professional Development Organization (2024 – now); Committee Member, Yale Physics Graduate Student Advisory Committee (2022 – 2023).